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Listening to active planetary surfaces

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Images reveal where planetary surfaces change, whereas seismic and ground-motion records can show when and why. Treating seismometers as environmental sensors, alongside cameras, meteorological packages and orbital repeat imaging, would make active surfaces a central target of future missions.

A dynamic view of planetary surfaces

Planetary surfaces have usually been read through photons. Reflected sunlight, thermal emission, laser altimetry and radar echoes have made it possible to map dunes, landslides, impact craters, dust-devil tracks, tectonic scarps and icy terrains across the Solar System. This view is extraordinarily powerful because it is spatially extensive and repeatable, but it is also incomplete: images often show what a surface looked like before and after an event, but not the mechanical perturbation that produced the change.

The next step is to add phonons to photons and make surface dynamics audible. The Seismic Experiment for Interior Structure (SEIS) on NASA's InSight Mars lander showed that a planetary seismometer is not only a probe of deep structure but also an environmental sensor. The SEIS recorded marsquakes, impacts, atmospheric pressure loading, wind-driven noise, convective vortices and the shallow response of the regolith to atmospheric forcing[1–3]. These signals are not simply noise to be removed before doing interior science. They are measurements of how the atmosphere, surface, shallow subsurface and interior couple mechanically.

This observation could motivate a shift in planetary surface science. Planetary environmental seismology should be treated as a component of geomorphology and geophysics, not as a byproduct of interior seismology. Its aim is to use seismic and ground-motion records, together with orbital and in situ observations, to constrain the processes that make planetary surfaces change. In this framework, photons identify where and how the surface changed, whereas phonons constrain when it moved, what loaded it, how waves propagated to it and which threshold, if any, was crossed. Not all environmental signals are failures: dust devils, for example, load the ground and mobilize dust without necessarily rupturing a surface. The broader target is mechanical coupling, from reversible loading to irreversible sliding, cracking or collapse.

By linking images and seismology, we can learn and infer much more (Fig. 1). A dust-devil track, a fresh crater, a boulder trail, a granular avalanche or an ice-shell crack is the endpoint of a chain connecting a source, environmental loading, wave propagation, near-surface amplification or attenuation, and the material response. Repeated imaging can detect the endpoint of that chain; seismology and meteorology can constrain its dynamics. The opportunity is joint photon–phonon inference.

Planetary playground

Mars already provides the clearest testbed. Around the InSight lander, newly formed dust-devil tracks could be compared with pressure, wind and seismic records of convective vortices[4,5]. A surface albedo feature could therefore be interpreted with the atmospheric vortex that loaded the ground and mobilized dust. Meteoroid impacts provide a complementary case: fresh craters seen from orbit can be linked to seismic signals, blast zones, secondary impacts and impact statistics[6–8]. The surface is not only the place where impacts happen; it becomes a distributed record of their mechanical aftermath.

Mass wasting shows why the dynamic constraint matters. Many Martian slope streaks are now interpreted as dry dust avalanches rather than merely albedo markings[9]. After the S1000a impact event and the S1222a marsquake, repeat orbital images were used to test whether dust-avalanche activity increased in ways compatible with seismic triggering[10]. In such cases, landforms behave as sparse threshold sensors: if a slope failed after a known event, while nearby slopes did not, the pattern contains information on ground motion, site susceptibility and local failure thresholds.

On the Moon there is no atmosphere to obscure the mechanical interpretation. Boulder trails, landslides and dry granular flows can record impacts, moonquakes, thermal degradation or preconditioned slope failure. Recent work has linked lunar boulder avalanches to shallow moonquakes, whereas global analyses show that impacts and crater-wall degradation can also dominate rockfall populations[11–13]. The lesson is not that every boulder trail has a seismic origin. It is that morphology alone is often insufficient to separate trigger, precondition and propagation effect.

Titan extends the argument to an icy, organic-rich world with a dense atmosphere. The Dragonfly lander will not carry a global geophysical network, but it will place seismic, meteorological and imaging observations inside a mechanically active environment. Modelling of tidal ice-cracking events shows that detectability depends on source geometry, ice-shell structure, attenuation, instrument response and environmental noise[14]. The same noise that complicates the detection of icequakes may also carry information about boundary-layer dynamics, surface–atmosphere coupling and the shallow mechanical state of organic sediments and ice.

From payloads to observatories

This is why mission design matters. A seismometer alone is powerful, but ambiguous. A seismometer operated with pressure, wind and temperature sensors, local imaging, topography and orbital repeat coverage becomes a process observatory. The immediate opportunity is not only to add new instruments to missions whose payloads are already fixed. It is to operate and archive existing geophysical packages with surface-process questions in mind. The Farside Seismic Suite, the South Pole Seismic Station and the Chang'e-7 seismograph on the Moon and Dragonfly on Titan can all be used to plan synchronized environmental monitoring, targeted local imaging, repeat orbital searches for new landforms, careful characterization of lander noise and open time registration between seismic, atmospheric and image data.

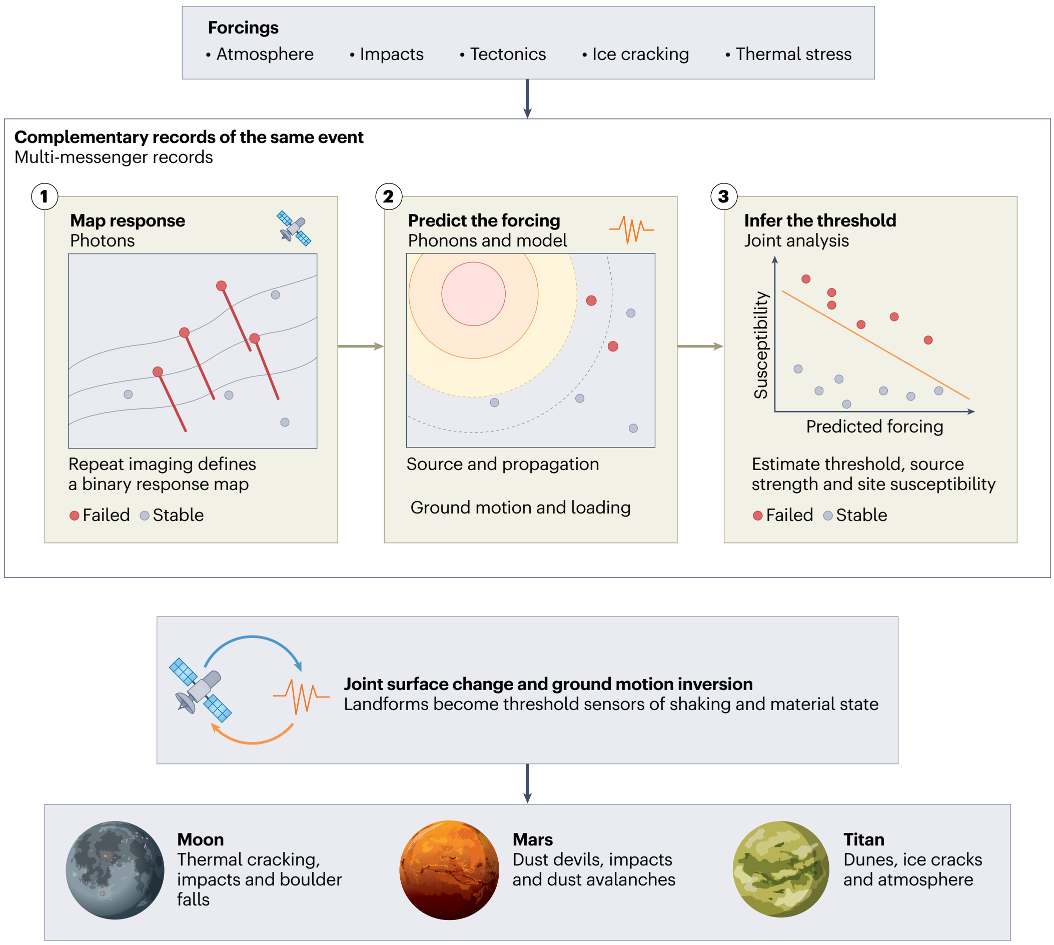

Fig. 1 | Planetary surfaces as photon–phonon observatories. Surface-active forcings such as atmospheric vortices, impacts, tectonic slip, ice cracking and thermal stress can produce both observable landform change and measurable ground motion. Imaging, topography, thermal and radar observations document the spatial response, whereas seismic, compliance and environmental-noise records constrain timing, forcing, propagation and material thresholds. Joint interpretation turns landforms into sparse threshold sensors of the mechanically defined planetary critical zone.

The same argument should guide the next payload calls and mission concepts. Later Artemis and commercial lunar payload opportunities could treat seismometers as monitors of impacts, thermal cracking, regolith scattering and astronaut- or rover-induced ground motion, not just as probes of the lunar interior. A future Mars network should be placed and operated so that active slopes, dust-rich terrains, impact-prone regions and tectonic structures are part of the target selection. Icy-world lander concepts should consider seismometers not only as tools for ocean detection, but also as sensors of cracking, tides, atmospheric forcing and surface mobility. In all cases, the requirement is the same: plan to capture the environmental context before launch rather than reconstruct it after an unexpected signal has been found.

Operationally, this means treating the surface itself as an instrument with uneven sensitivity. Avalanches, boulder falls, dust-removal patches and fracture openings are not continuous sensors, but they encode whether local thresholds were exceeded. Their absence is also informative when the preservation conditions and image coverage are known. Combined with even a single seismic station, such thresholded observations can reduce ambiguity in source location, attenuation and site response; with several stations, geomorphic change could be assimilated, together with waveform and environmental data.

The same logic changes how planetary surfaces are conceptualized. They are not only archives of past events. They are mechanically active interfaces where regolith, ice, atmosphere, impacts, tides and internal activity interact. Planetary environmental seismology therefore provides a way to probe a mechanically defined planetary critical zone. Unlike the terrestrial critical zone, this interface need not be water- or biosphere-dominated. Its defining property is dynamic coupling: exterior forcing and interior activity are transmitted through near-surface materials and become observable as deformation, shaking, cracking, sliding, dust lifting or failure.

The practical goal is direct. When an image shows a new crater, dust avalanche, boulder fall or fracture, we should ask not only what changed on the surface, but also what force acted, when it acted, how it propagated and what material response it produced. Conversely, when a seismometer records an impact, vortex, quake or ice crack, the question should not only be what it says about the interior, but also what surface process it may have triggered or revealed. Photons made extraterrestrial landscapes visible; seismology can make their dynamics measurable. The task now is to listen to active planetary surfaces as deliberately as we look at them.

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Competing interests

The authors declare no competing interests.